

Project Title
Symptom-to-Diagnosis Matcher

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

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ABSTRACT

Symptom-to-Diagnosis Matcher is a Java-based decision-support engine that maps free-text patient symptoms to likely medical conditions using a structured medical knowledge base. The system first normalizes the entered symptom descriptions and converts them into standard symptom terms. It then compares the identified symptoms with symptoms associated with different candidate diagnoses and assigns transparent scores based on the strength of each match. The candidate diagnoses are ranked according to their scores to identify the most relevant possibilities. The system also recommends a minimal set of diagnostic tests that can best distinguish between the top-ranked conditions. The project demonstrates the application of data structures and algorithms such as hashing, searching, sorting, scoring, ranking, and set-based selection. The system is intended as an educational decision-support prototype and does not replace professional medical diagnosis.

Introduction

Medical symptoms can be described in different ways using natural language. For example, users may enter terms such as "high temperature," "feeling hot," or "fever," even though they may refer to the same symptom. This makes direct matching with a medical knowledge base difficult.

The **Symptom-to-Diagnosis Matcher** addresses this problem by normalizing free-text symptoms and matching them with a structured medical knowledge base. The system calculates a transparent score for each candidate diagnosis based on matching symptoms and ranks the possible conditions.

The system further analyzes the top-ranked diagnoses and recommends a minimal set of diagnostic tests that can help distinguish between them. Java data structures such as **HashMap, HashSet, ArrayList, and PriorityQueue**, along with searching, sorting, and scoring algorithms, can be used to efficiently implement the system.

The project demonstrates how Data Structures and Algorithms can be applied to build an explainable decision-support system.

Software and hardware requirements

Software Requirements

- **Operating System:** Windows
- **Programming Language:** Java
- **Java Version:** JDK 17 or above
- **IDE:** Visual Studio Code
- **Database / Knowledge Base:** MySQL
- **Frontend:** Java Swing
- **Version Control:** GitHub

Hardware Requirements

- **Processor:** Intel Core i3 or equivalent and above
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 500 MB free space

- **Display:** Standard monitor
- **Input:** Keyboard and mouse
- **Internet:** Required for downloading development tools and optional knowledge-base resources

Course Instructor